

**BRAC University**  
**Dept. of Computer Science and Engineering**  
**Spring 2025**  
**CSE331-04**

**Time: 20 minutes**

**Marks: 20**

Name:

ID:

Section:

1. Convert the following regular expression into context free grammar. [8]

$$(a+bc)cbb(cc+b)^* + aa$$

2. Give a context-free grammar that generates the language [6+6=12]

a.  $L = \{ w \in \{0, 1, 2\}^* : w = 1^i 2^j 0^k, \text{ where } k = 2i + 3j, i, j \geq 0 \}$

b.  $L = \{ w \in \{a, b\}^* : w \text{ starts with odd number of 'b's'} \}$

$$1. S \rightarrow XYZ | aa$$

$$X \rightarrow a | bc$$

$$Y \rightarrow cbb$$

$$Z \rightarrow wZ | \epsilon$$

$$w \rightarrow cc | b$$

$$2. a) 1^i 2^j 0^k \quad k = 2i + 3j$$

$$1^i 2^j 0^{2i+3j}$$

$$S \rightarrow 1S00 | X$$

$$X \rightarrow 2X000 | \epsilon$$

$$b) (bb)^*b + (bb)^*ba(a+bb)^*$$

$$S \rightarrow Max | M$$

$$M \rightarrow bbM | b$$

$$X \rightarrow aX | bX | \epsilon$$